

NKABOM RESEARCH PROJECTS · ACCRA, GHANA

CARBON CLARITY

Three projects. One mission: a cleaner stove, trusted data, and carbon-credit income that reaches the women who smoke fish.

Behind every smoked fish in Ghana is a woman working inside the smoke.

THE PROBLEM WE ARE SOLVING

Most of Ghana's fish is smoked by women over open wood fires, in Traditional/kilns designed in 1969. They breathe the smoke, buy firewood every day, and cannot afford cleaner stoves. Carbon finance could pay for the switch, but there is no trusted field data to prove the savings, and carbon income rarely reaches the women who make the change.

Her health
Smoke damages eyes and lungs and their backs due to bending while using it. Fish smokers breathe about 2.7× more carbon monoxide than other women.

Her income
Firewood is a daily cash cost on top of fish and kiln rent. Improved ovens cost many times a traditional kiln.

Our forests
Open fires waste wood fuel, adding pressure on the standing trees and releasing avoidable carbon emissions.

Her fair share
No verified data, so no carbon credits; and most carbon projects never pay the women themselves.

70–80%
of Ghana's fish is smoked

>80%
of processors still use the 1969 Chorkor kiln

≈700
improved ovens in use (2021), for 30,000+ processors

≈GHS 10,000
reported price of an improved oven (2024)

Sources: Asiedu et al. 2018; CSIR-FRI 2022; Coastal Resources Center 2021; Weyant et al. 2022 (Environ. Health Perspect.); Asiedu 2024 (MyJoyOnline).

OUR AIMS

Over the page: our three-phase plan →

1 Understand
Document how fish smoking works today: kilns, fuel, smoke, health and money.

2 Prove
Test the traditional smoker (kiln) against an improved stove, side by side, in the field.

3 Measure
Collect sensor data on fuel and cooking that carbon auditors will trust through IoT sensors.

4 Reward
Turn verified savings into carbon credits, with income paid to the women by mobile money.

OUR THREE-PHASE PLAN

PHASE 1

IN PROGRESS

Understand & compare

baseline fieldwork at James Town (kilns, fuel, smoke, costs)

- Working paper on the baseline (Dec 2026)
- Side-by-side study: traditional kiln vs improved stove

OUTPUT Evidence and a design blueprint

PHASE 2

NEXT

Design & measure

- Refine the stove with the women who use it
- Install IoT sensors on stoves in the field
- Collect fuel and cooking data to carbon-audit standard

OUTPUT A proven stove and trusted data

PHASE 3

LATER

Reward & scale

- Pilot the benefit-sharing model
- Turn verified savings into carbon credits
- Pay credit income to the women by mobile money
- Implementation of the stove

OUTPUT Income in women's hands

THREE PROJECTS UNDER ONE UMBRELLA



THE CLEANSTOVE

Improved smoking stove

A side-by-side comparison of the traditional kiln and an improved stove becomes the blueprint for a design built for the women who use it.



THE DATA MACHINE

IoT sensors in the field

Sensors capture fuel and cooking data directly from the field, built to meet the evidence standards carbon auditors require.



THE CARBON CREDIT MONEY

Benefit-sharing model

Designed so carbon-credit income reaches the women directly, by mobile money. Model in development.

WHY IT MATTERS

Improved stove › **Less firewood burned** › **Less pressure on trees** › **Lower carbon emissions** › **Credits paid to HER**

Target: women spend up to 50% less on firewood, to be confirmed by the comparative study. That money stays with her.

WHERE WE ARE

✓ Phase 1 underway: James Town baseline done (Oct 2026)

→ Now: baseline paper + comparative stove study

→ Then: Phase 2 sensors · Phase 3 benefit-sharing

JOIN US

Photos: Carbon Clarity team, James Town, 2026. Stove, sensor and benefit-sharing components are planned and not yet deployed.